



Cortical EEG slowing in resting-state Parkinson's disease

Final presentation: model, code, and findings

PD EEG group · Rui, Jan, Melissa, Friedrich · 26.06.2026

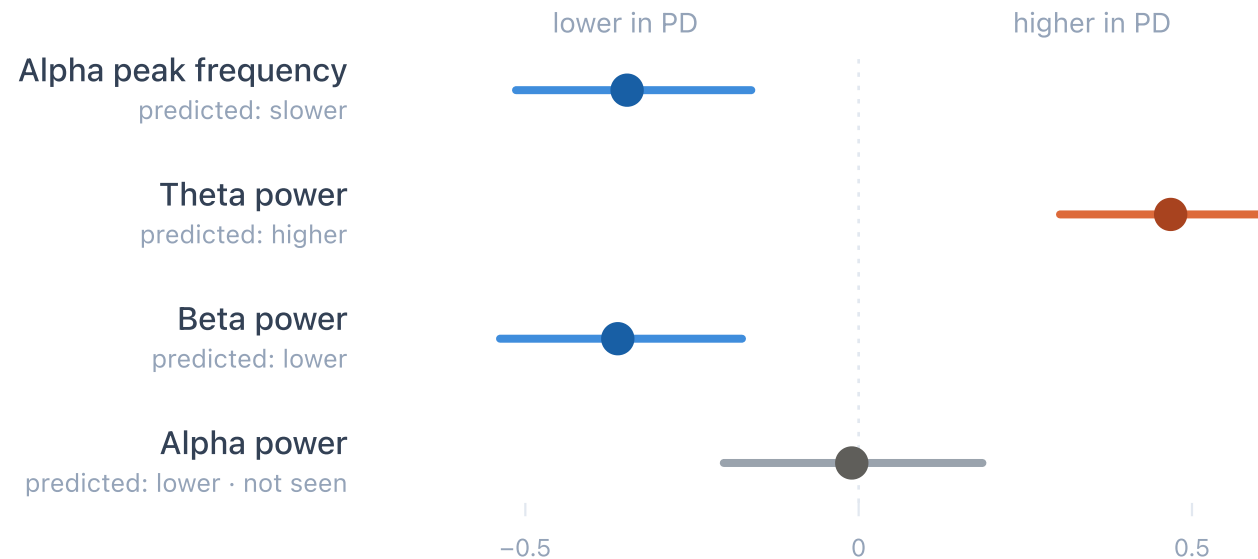
Hypotheses

Compared to age-matched healthy controls, Parkinson's patients at rest show:

- **slowing of the alpha peak frequency** (8 to 13 Hz)
- **increased theta-band power** (4 to 8 Hz)
- **decreased cortical beta-band power** (13 to 30 Hz)

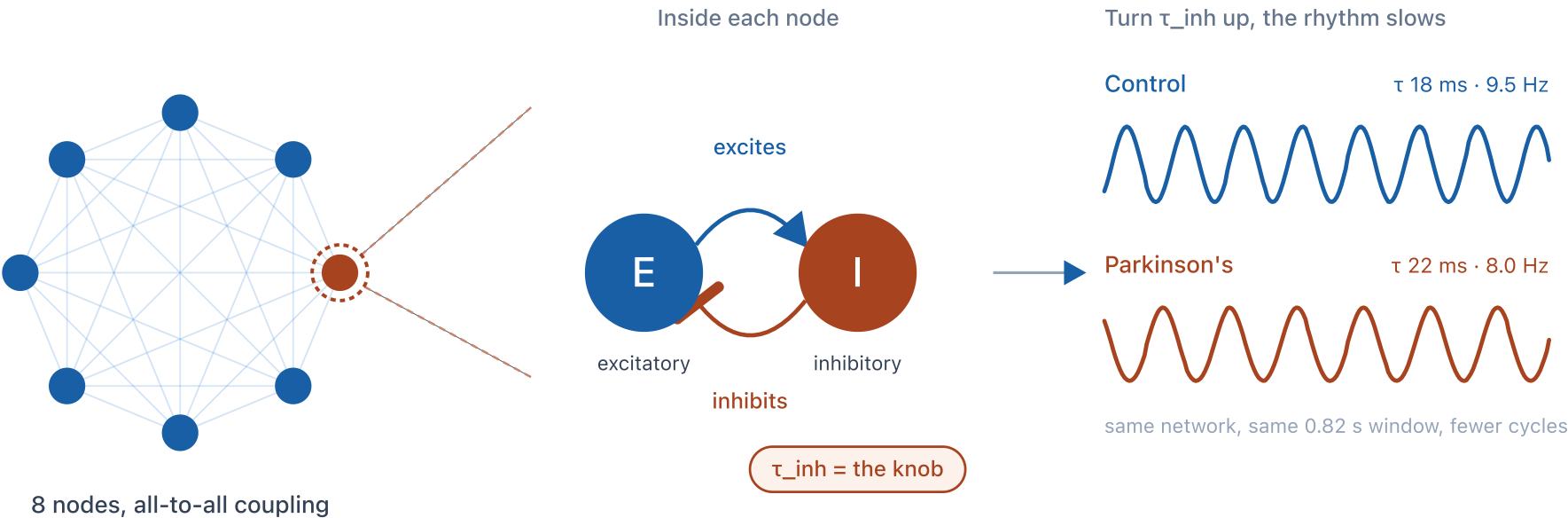
How Parkinson's changes the resting EEG

effect size per feature, with 95% confidence interval



The model: a Wilson-Cowan network

8 coupled cortical nodes, each an excitatory-inhibitory loop. τ_{inh} is the only knob we turn.



The final adaptation from healthy to patient is a single parameter: a longer inhibitory time constant τ_{inh} , about **+3.25 ms** in PD.

Code & repository

THE PIPELINE



02 builds the tau-to-peak curve that 03 inverts · 04 tests the data directly · 06 validates the fit

THE CODE

1

Fit each subject
src/fitting.py

`tau = invert_peak_curve(alpha_peak, ...)`
tau from each subject's alpha peak

2

Build the model
src/model.py

`model = build_network(..., tau_inh=tau)`
one number = healthy vs patient

3

Compare groups
src/stats.py

`compare_groups(..., alternative="less")`
one-sided test: PD tau higher

every result is one script run

```
bash
$ python scripts/05_compare_tau.py
Control 18.25 → PD 21.50 ms diff +3.25 p = 0.0025 ***
→ wrote results/figures/compare_tau.png
$ python scripts/06_recover_tau.py
22 known tau recovered bias -0.10 max err 1.75 ms
→ wrote results/tau_recovery.csv
```

THE REPOSITORY

pd-eeg

main

scripts/
01_extract_features.py
02_simulate_network.py
03_fit_per_subject.py
04_compare_features.py
05_compare_tau.py
06_recover_tau.py

src/
results/
fitted_tau.csv
group_comparison_tau.csv
tau_recovery.csv

figures/
slides/

numbered pipeline
features · model · fitting · stats
every run writes here
compare_tau.png · tau_recovery.png · ...
decks + notes

6 scripts · 4 modules · 46 commits · 149 subjects

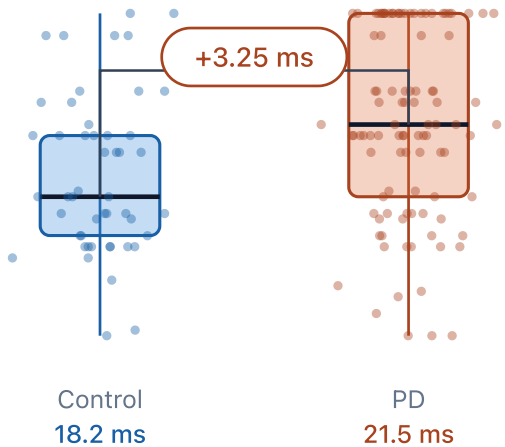
Everyone merges into main



Does the model separate patients from controls?

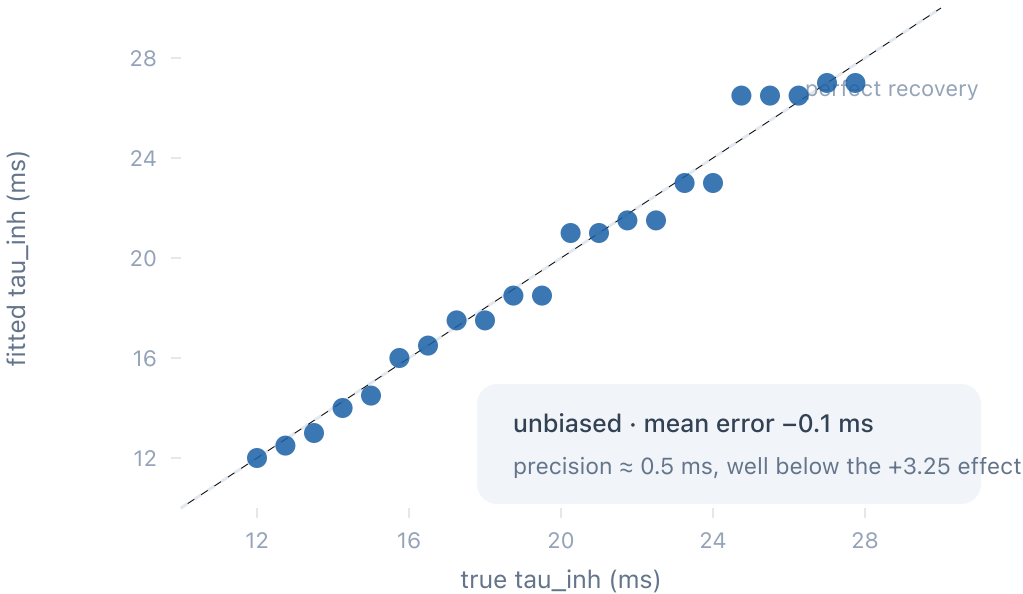
Fitted inhibitory time constant

the model parameter we change for a patient
fitting recovers tau to ± 0.5 ms, far below this group gap



Validation: recovering a known parameter

simulate at a known tau, fit it back



Fitted τ_{inh} is **+3.25 ms** higher in PD (one-sided Mann-Whitney, $p \approx 0.0002$). The recovery test shows this is a real separation, not an artefact of the fitting.

Outlook and open challenges

CURRENT CHALLENGES

- One parameter (τ_{inh}) reproduces the alpha slowing, but not the theta and beta changes
- The fit re-expresses the data-level effect, rather than adding independent evidence

FUTURE DIRECTIONS

- Fit more than the peak (full spectrum or functional connectivity), on a real connectome
- Relate fitted τ to clinical scores and to known inhibitory (GABAergic) changes in PD

What we took away

Rui [your biggest learning]

Jan [your biggest learning]

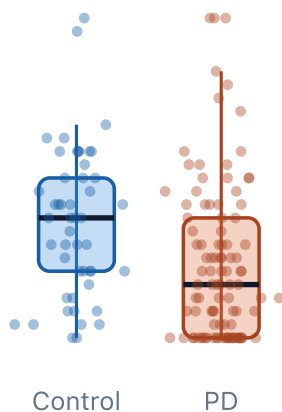
Melissa [your biggest learning]

Friedrich [your biggest learning]

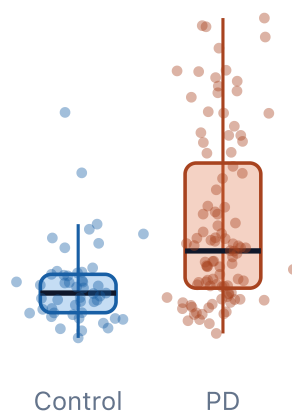
Appendix: per-subject distributions

PD vs Control: per-subject EEG features

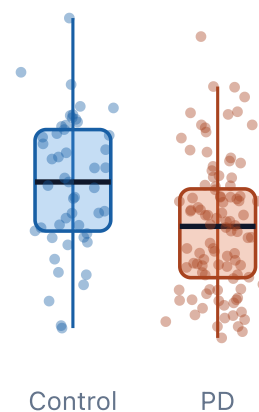
Alpha peak frequency
slower in PD



Theta power
higher in PD



Beta power
lower in PD



Alpha power
unchanged

